

PATENT APPLICATION ARCHITECTURAL BLUEPRINT

Invention Title: System and Method for Near-Infrared Photothermal Phase-Change Drug Delivery with Integrated Chemiluminescent Telemetry and Advanced Clearance Management

System Designation: Adriel Shamer Willis: RGBY System

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1. Abstract

An advanced, light-triggered non-mechanical drug delivery platform is disclosed for the regulated administration and active elimination of targeted pharmaceuticals. The platform incorporates an ingestible solid-state electronic circuit or a biocompatible matrix containing macromolecular lipid dendrimers. These are co-loaded with an active pro-drug, an upconverting photothermal nano-agent, a low-boiling-point phase-change liquid core, and a chemiluminescent reporter matrix. Upon wireless trigger from an external wearable terminal, near-infrared (NIR) light is converted into localized high-energy wavelengths, inducing a non-mechanical phase-change membrane disruption to achieve instant bioavailability. Active systemic toxicity mitigation is governed by light-activated molecular hinges and cleavable polymeric chains that selectively force ultra-rapid biliary and renal clearance on command.

2. Background of the Invention

Modern drug delivery systems are heavily constrained by physical mechanics when engineered at a micro-scale. Moving parts, micro-compressors, or pressurized pneumatic valves introduce fatal friction limits (stiction) and require unsustainable electric currents. Furthermore, high-potency neurological compounds often present severe pharmacokinetic risks, including long systemic half-lives, unstable plasma concentration curves, and accumulated organ toxicity. Prior art systems fail to provide a mechanism that can turn an active compound on and off dynamically while simultaneously accelerating organ-level elimination paths without inducing vascular shear stress.

3. Detailed Description of the Preferred Embodiments

A. Stimulus-Responsive LNP Architecture and Structural Synthesis

The *Adriel Shamer Willis: RGBY System* synthesizes macromolecular Poly(amidoamine) (PAMAM) Generation-5 (G5) dendrimers shielded via surface-bound Polyethylene Glycol (PEGylation) chains to guarantee complete immune system stealth. To enable on-demand disassembly, the core polymer branches are linked via photocleavable **ortho-nitrobenzyl (ONB) groups**. The standard chemical synthesis comprises the covalent conjugation of a modified 4-[4-(1-hydroxyethyl)-2-methoxy-5-nitrophenoxy]butyric acid linker to the

amine termini of the G5 dendrimer, followed by esterification of the active pharmaceutical pro-drug onto the remaining hydroxyl nodes. Indocyanine Green (ICG) molecules are embedded into the lipid outer shell to act as near-infrared optical transducers.

B. Upconversion Phase-Change Mechanics

Rather than relying on mechanical fluid pumps, the application of external near-infrared light (700–900 nm) passes harmlessly through tissue boundaries where it is absorbed by the internal ICG matrix. This optical energy induces localized photothermal heating up to a critical phase-change threshold of 41°C to 43°C. This microscale thermal shift triggers an instantaneous liquid-to-gas transition of an internal perfluorocarbon (PFC) fluid core. The rapid volumetric expansion ruptures the protective LNP boundary, discharging the pro-drug payload directly into local vascular beds without generating systemic physical gas embolisms.

C. Advanced Excretion and Clearance Mathematics

To actively reverse systemic drug accumulation, the system leverages a dual-path elimination engine. Under a secondary wavelength trigger (400–420 nm upconverted photons), the ortho-nitrobenzyl linkers undergo rapid photo-cleavage, shattering the 8–10 nm dendrimer shield into fragments smaller than 3 nm. This allows immediate clearing via the kidney glomerulus. The mathematical clearance velocity is governed by the following systemic equation:

$$CL_{\text{renal}}(t) = GFR \cdot [1 - \exp(-k_{\text{photo}} \cdot I_{\text{optical}} \cdot t)] \cdot \exp(-\lambda_{\text{metabolic}} \cdot t)$$

Where GFR represents the baseline glomerular filtration rate, k_{photo} represents the specific ONB photolytic cleavage constant, I_{optical} is the optical power density transmitted from the external sensor, and $\lambda_{\text{metabolic}}$ represents independent baseline plasma clearance pathways. Simultaneously, lipophilic components are bound to photosensitive biliary tags; activation of the 420 nm switch flips an azobenzene core, exposing a cholic-acid marker that forces transport proteins (OATPs) to pump the active drug directly into the bile duct.

D. Chemiluminescent Closed-Loop Feedback

Real-time dosing telemetry is established via a co-encapsulated firefly-derived luciferin-luciferase reporter matrix. Upon photothermal membrane disruption, the matrix undergoes Bioluminescence Resonance Energy Transfer (BRET), emitting a cold, non-thermal light signal back through the tissue. An external smart wrist wearable quantifies this optical flux to track exactly how many milligrams have crossed into the biological environment, modulating laser energy automatically to match target therapeutic parameters.

4. Patent Claims

What is claimed is:

1. A light-modulated, non-mechanical drug delivery and clearance platform comprising: an ingestible or implantable solid-state electronic device; a formulation comprising poly(amidoamine) (PAMAM) dendrimers conjugated to a therapeutic agent via an ortho-nitrobenzyl photo-cleavable linker; a near-infrared absorbing photothermal transducer; and an external wearable module including an optical sensor configured to

execute closed-loop dosing calculations based on a chemiluminescent telemetry feedback loop.

2. The platform of claim 1, wherein said poly(amidoamine) dendrimers are structurally encapsulated within surface-bound polyethylene glycol (PEG) chains configured to prevent immune system recognition and enzymatic degradation prior to an optical activation event.
3. The platform of claim 1, wherein said ortho-nitrobenzyl photo-cleavable linkers are configured to undergo high-velocity photolytic cleavage upon exposure to an upconverted optical wavelength between 400 nm and 420 nm, shattering said dendrimer structure into molecular fragments smaller than 3 nanometers to facilitate immediate glomerular filtration via the kidneys.
4. The platform of claim 1, further comprising a secondary clearance pathway wherein the therapeutic agent is linked to a photosensitive biliary targeting conjugate containing an azobenzene core hinge, wherein an upconverted optical prompt causes shape-shifting structural conformation that exposes a cholic-acid excretion tag to prompt direct hepatic-biliary elimination.
5. The platform of claim 1, wherein the mathematical clearance velocity of the shattered fragments through the renal filtration boundary is directly proportional to the optical power density (I_{optical}) delivered by the external wearable module matching the formula: $CL_{\text{renal}}(t) = GFR \cdot [1 - \exp(-k_{\text{photo}} \cdot I_{\text{optical}} \cdot t)] \cdot \exp(-\lambda_{\text{metabolic}} \cdot t)$.

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